



counting sort 12

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Complete the countingSort function.

countingSort has the following parameter:

int arr[n]: an array of integers Returns Nothing. Print the sorted array with each element separated by a space.

Input Format

The first line contains an integer n, the number of elements in the array. The second line contains n space-separated integers. 10 11 3 2 1 4 5 6 4 3

Constraints

$1 \leq n \leq 100000$ $0 \leq arr[i] < 100$

Output Format

1 1 1 2 3 3 4 4 5 6

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Language

C++11



```
1  #include <cmath>
2  #include <cstdio>
3  #include <vector>
4  #include <iostream>
5  #include <algorithm>
6  using namespace std;
7
8
9  int main() {
10     int n;
11     cin >> n;
12     vector<int> arr(n);
13     for(int i=0; i<n; i++)
14     {
15         cin >> arr[i];
16     }
17     vector<int> count(100,0);
18     for(int i=0; i<arr.size(); i++)
19     {
20         count[arr[i]]++;
21     }
22     for(int i=0; i<100; i++)
23     {
24         while(count[i]>0)
25         {
26             cout << i <<" ";
27             count[i]--;
28         }
29     }
30     return 0;
```

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```
10
11 1 5 1 2 6 4 3 1 2
```

//

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Input (stdin)

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```
1 10
2 11 1 5 1 2 6 4 3 1 2
```

Your Output (stdout)

```
1 1 1 1 2 2 3 4 5 6 11
```